

# AI ASSIGNMENT - 3

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# INTRODUCTION

**Minimax Algorithm:** A decision-making algorithm used in two-player zero-sum games, where each player alternates turns.

**Purpose of Comparison:** To analyze the differences between Minimax pruning (Alpha-Beta Pruning) and the basic Minimax approach.

# MINIMAX - AN OVERVIEW

## Working Principle:

- Maximize the score for the maximizing player.
- Minimize the score for the minimizing player.

**Tree Structure:** A game tree where each node represents a game state, and edges represent moves.

**Outcome:** The algorithm explores all possible moves to determine the best strategy for each player

**Alpha-Beta Pruning:** Introduces two parameters, Alpha and Beta, to "prune" branches that won't be selected.

- **Alpha:** The best value that the maximizing player can guarantee.
- **Beta:** The best value that the minimizing player can guarantee.

# PERFORMANCE IN GENERAL

- **Non-Pruning:**
  - Time Complexity:  $O(b^d)$
  - Space Complexity:  $O(b \cdot d)$
- **With Pruning:**
  - Best Case Time Complexity:  $O(b^{\{d/2\}})$
  - Space Complexity: Still  $O(b \cdot d)$ , but the algorithm explores fewer nodes.

# COMPARISONS IN GENERAL

## Comparison - Efficiency

- **Minimax (No Pruning):**
  - **Time-consuming**, as it evaluates all nodes.
  - High space complexity due to storing the full tree.
  - Slow in large game trees.
- **Minimax with Alpha-Beta Pruning:**
  - Significantly reduces the number of nodes evaluated (best case  $O(bd/2)$   $O(b^{d/2})$   $O(bd/2)$ ).
  - Can prune entire subtrees early, improving efficiency.
  - Potential for **much faster decision**.

## Comparison – Search Space

- **Minimax (No Pruning):** Searches the entire tree.
- **Minimax with Pruning:** Avoids unnecessary subtrees, focusing only on relevant part

# MINIMAX(NO PRUNING)



# MINIMAX (WITH ALPHA-BETA PRUNING)



# WHAT'S THE DIFFERENCE?

Alpha-beta pruning is effectively an optimisation on minimax, and since minimax as algorithm is **deterministic** - it will always return only one result from one starting state, this means the result looks the same, the difference is how **fast** each algorithm finds the node (the next move).

```
atreya@atreya-Dell-G15-5511:~/Downloads$ python3 benchmark.py
Minimax move: g1h3
AlphaBeta move: g1h3
-----
Minimax time: 0.2531s | Nodes: 0
AlphaBeta time: 0.0137s | Nodes: 0
```

As you can see, for the same node (move), the alpha-beta pruning yields a speed-up of **18.4x** !



# CONCLUSION

Minimax with Pruning offers a substantial performance boost by reducing the number of nodes evaluated.

Non-Pruning is straightforward but inefficient for large game trees.

Alpha-Beta Pruning: Essential for practical use in games with large state spaces like Chess and Checkers.